NAME: ABDULLAH **ROLL NO: 044**

CLASS: BSAI **SECTION: 3A**

LAB TASK NO: 4

Question:1) A* First Three Expansions

Formula:

f(n)=g(n)+h(n)

g = Actual cost

h = Estimated cost

f = g + h

Step	Node	g	h	f
1	S	0	5	5
2	A	1	4	5
3	C	3	2	5

Calculations:

S 0 + 5 = 5

A 1 + 4 = 5

C 3 + 2 = 5

G 6 + 0 = 6

Frontier:

After S A 5, B 8

After A C 5, B 8

After C G 6, B 8

Final Path: S-A-C-G

Total Cost = 6

Question:2) A* vs UCS

```
import heapq
graph = {
```

```
    'S': [('A', 1), ('B', 4)],
    'A': [('C', 2)],
    'B': [('G', 5)],
    'C': [('G', 3)],
    'G': []
}
h = {'S': 5, 'A': 4, 'B': 4, 'C': 2, 'G': 0}
def search(heuristic):
    q = [(heuristic['S'], 0, 'S', ['S'])]
    best = {'S': 0}
    count = 0
    order = []
 while q:
        f, g, node, path = heapq.heappop(q)
     if g != best[node]:
            continue
order.append(node)
if node == 'G':
            return path, g, order, count
   count += 1
for child, cost in graph[node]:
            new_g = g + cost
 if new_g < best.get(child, float('inf')):
                best[child] = new_g
                heapq.heappush(q, (
                    new_g + heuristic[child],
                    new_g, child, path + [child]))
return [], 0, order, count
print("A*:", search(h))
zero_h = {node: 0 for node in graph}
```

```
print("UCS:", search(zero_h))
```

Results:

Detail	A*	UCS
Visit order	S,A,C,G	S,A,C,B,G
Path	S-A-C-G	S-A-C-G
Cost	6	6
Expansions	3	4

Question:3) Heuristic Overestimate

Change only:

h(C): 2-10

C true remaining cost is 3, so 10 is an overestimate.

Modified Results:

Step	Node	g	h	f
1	S	0	5	5
2	A	1	4	5
3	B	4	4	8
4	G	9	0	9

GITHUB LINK:

https://github.com/abdullahrehman957-lgtm/AI-LAB-TASK